

SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 2.7: Surface Area and Volume of Solids — Teacher Reference (Pre-filled)

SUMMARY TABLE: CORE MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 2.7: Surface Area and Volume of Solids
Driving Question	DRIVING QUESTION: Why does the same amount of metal sheet make containers that hold very different amounts of grain — and how can a Kenyan artisan use mathematics to design the best solid shape for storage or construction?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring Phenomenon & Surface Area of Prisms: Same Metal, Different Grain	We observed two closed containers at Mzee Kiplagat's posho mill in Eldoret, a cylindrical drum and a square-based box, both made from 30,800 cm ² of iron sheet and both about 1 m tall, yet the drum holds more maize. We also measured the faces of a 5 cm x 4 cm x 3 cm rectangular prism (TSA 94 cm ²) and a 3-4-5 cm triangular prism of length 10 cm (TSA 132 cm ²).	The total surface area of a prism is the sum of the areas of all the faces in its net: $TSA = 2 \times \text{base area} + \text{perimeter of base} \times \text{length}$. The box (50 cm x 50 cm x 129 cm) has $TSA = 5,000 + 25,800 = 30,800 \text{ cm}^2$. Surface area measures the material covering a solid, and volume measures the space inside, so they are different quantities.	Wanjiru used 30,800 cm ² of sheet for the box, which agrees with our calculation, so the same amount of material is used for both containers. But knowing the surface area does not tell us how much maize a container holds, because capacity depends on volume. We still need to find the drum's surface area and both volumes to explain why the drum holds more; this is now on our DQB.
Lesson 2 Surface Area of Pyramids: Slant Height, Vertical Height and Roof Cover	We built a square-based pyramid (base 6 m, vertical height 4 m, scale model) and a regular triangular pyramid (tetrahedron, edges 6 cm). We measured the vertical height and the slant height and saw that the slant height (5 m) is longer than the vertical height (4 m). We saw that the triangular faces need the slant height.	Slant height (l) is measured along a triangular face; vertical height (h) goes straight down from the apex to the base. We can find l using Pythagoras ($3^2 + 4^2 = 5^2$). Pyramid $TSA = \text{base area} + \text{sum of the areas of the triangular faces}$. The roof area $= 4 \times \frac{1}{2} \times 6 \times 5 = 60 \text{ m}^2$, and the tetrahedron has TSA of about 62.35 cm ² .	Wanjiru can use the net and sum-of-faces method to find how much iron sheet any flat-faced solid needs, and she must use the slant height for the triangular faces. Surface area only measures the covering material, not the space inside, so it cannot yet explain why the drum and box with the same 30,800 cm ² hold different amounts of maize. We need to study volume to answer that.
Lesson 3 Surface Area of Cones and Frustums: From Sector Nets to Turkana	We observed that the curved surface of a paper cone, when cut along its slant edge and flattened, forms a sector of a circle. The straight edge of the sector equals the slant	We learned that the curved surface area of a cone $= \pi r l$, and the total surface area $= \pi r(r + l)$. The frustum's curved surface area $= \pi(R + r)l$, which equals the big cone	We can explain that an artisan like Wanjiru finds the sheet needed for cones and frustums by unrolling them into sector nets and using $\pi r l$ or $\pi(R + r)l$, so surface area

Water Tanks	height l , and the arc length equals the circumference of the cone's base ($2\pi r$). We also observed that a frustum is a big cone with a smaller cone removed from the top, using the ice-cream cone ($r = 3.5$ cm, $l = 12$ cm), the conical tank top ($r = 1.4$ m, $l = 2$ m) and the Turkana frustum tank.	minus the removed cone. We learned that the ice-cream cone has a curved SA of 132 cm ² , the conical tank top needs 8.8 m ² of sheet, and the Turkana frustum needs 17.325 m ² (checked by $23.1 - 5.775$). We also learned to state whether an area is curved or total and to use slant height l , not vertical height h , in these formulae.	tells us how much material covers a solid. This only measures the material, not the space inside, so we still need volume to explain why Mzee Kiplagat's drum and box, made from the same 30,800 cm ² of sheet, hold different amounts of maize.
Lesson 4 Surface Area of Spheres and Composite Solids: Wrapping the Football, Decomposing the Container	We wrapped a football ($r = 10.5$ cm) in strips and found the skin covered about 4 circles of radius 10.5 cm. We saw that the Nairobi container is a cylinder with a hemisphere on top, with a hidden joined circle. We added up the parts of the drum and got 30,800 cm ² , the same as the box.	Sphere SA = $4\pi r^2$ (football 1,386 cm ²). Hemisphere curved SA = $2\pi r^2$. For composite solids we decompose into familiar parts, add only the outside faces, and leave out the hidden joined faces (Nairobi container 902 cm ²). Surface area is the amount of material covering a solid, not the space inside.	The drum and the box both use 30,800 cm ² of sheet, so surface area alone cannot tell Mzee Kiplagat which holds more maize. Surface area measures the material covering a solid, so we still need to find the space inside (volume) in Lessons 5 and 6 to explain why the drum holds more.
Lesson 5 Volume of Prisms and Pyramids: How Much Maize Can the Box Hold?	We filled the 5 x 4 x 3 cm and 3-4-5 triangular prism models with sand and both held 60 cm ³ . A pyramid with the same base and height as a prism had to be filled 3 times to fill the prism. The box from Eldoret measures 50 cm x 50 cm x 129 cm.	Volume of a prism = base area x height; volume of a pyramid = $\frac{1}{3}$ x base area x height, using the vertical height. Volume is measured in cm ³ or m ³ , and 1 litre = 1,000 cm ³ . The box holds 322,500 cm ³ = 322.5 litres, and the pyramid roof encloses 48 m ³ .	The box and drum use the same 30,800 cm ² of sheet, but surface area only measures the material covering a solid, not the space inside. The box volume of 322,500 cm ³ comes from base area x height, so equal surface area does not give equal volume. We still need the drum volume (Lesson 6) to see which container holds more maize.
Lesson 6 Volume of Cones, Frustums and Spheres: Resolving Why the Drum Holds More Maize	We observed that 3 cone-fuls of water or sand fill a cylinder with the same radius (7 cm) and height (12 cm), so the cone holds 616 cm ³ and the cylinder 1,848 cm ³ . We also saw the football ($r = 10.5$ cm) hold 4,851 cm ³ and the Turkana frustum tank hold 11.319 m ³ . The drum holds 404,250 cm ³ and the box 322,500 cm ³ , although both used 30,800 cm ² of sheet.	We learned that $V(\text{cone}) = \frac{1}{3}\pi r^2 h$, $V(\text{sphere}) = \frac{4}{3}\pi r^3$, and $V(\text{frustum}) = \text{big cone minus small cone}$. We learned that surface area measures the material covering a solid, while volume measures the space inside, so they are different quantities.	We explained that the drum holds 81.75 litres more maize than the box because, with the same surface area of 30,800 cm ² , the drum's shape (circular cross-section) encloses more space than the box's square cross-section. Shape, not only the amount of material, determines capacity, so Wanjiru can use mathematics to choose shapes that give more capacity for the same sheet.
Lesson 7 Composite Solids: Volume, Surface Area & Real-Life Applications	We observed that the hemispherical-top water tank (cylinder $r = 2.1$ m, $h = 4$ m) has a total surface area of 94.38 m ² and a volume of 74.844 m ³ (74,844 litres). We also observed that the Rift Valley silo (cylinder $r = 2.1$ m, $h = 6$ m with a cone roof) has a total surface area of 116.16 m ² and a volume of 96.096 m ³ .	We learned that a composite solid is split into known solids; for surface area we add only the outside faces (the joining circles are hidden), and for volume we add the volumes of the parts. Surface area (m ²) tells us the material or paint needed, and volume (m ³ or litres) tells us the capacity, so we choose the one that the question needs. 1 m ³ = 1,000 litres.	We explain that the drum and the box in Mzee Kiplagat's posho mill have the same surface area (30,800 cm ²) but different volumes (404,250 cm ³ and 322,500 cm ³) because shape, not just the amount of sheet, decides capacity. A Kenyan artisan therefore calculates both surface area and volume, as for the tank and silo, to choose a shape and dimensions that give the capacity needed for the least sheet and

			cost.
Lesson 8 Project Presentations, Model Building & Unit Consolidation: Why the Drum Holds More Than the Box	What we observed: Two closed containers made from the same 30,800 cm ² of iron sheet, a drum (r 35 cm, h 105 cm) and a box (50 cm x 50 cm x 129 cm), held different amounts of maize, and the groups' own container designs showed the same pattern: the shape changed the capacity.	What we learned: Surface area measures the material covering a solid and volume measures the space inside; the drum holds 404,250 cm ³ (404.25 litres) and the box 322,500 cm ³ (322.5 litres), so the drum holds 81.75 litres more; we can find both for prisms, pyramids, cones, frustums, spheres and composite solids.	What we can explain: For a fixed surface area, shape decides capacity because a circle encloses more area for its perimeter than a square, and a sphere is best of all, so artisans like Wanjiru use mathematics to choose shapes and dimensions that give the largest capacity for the least sheet metal, balanced against stacking and cost.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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